

F24-063-D-EDA

Project Team

Manahil Kamran 21I-2668
Ali Arfa 21I-2669

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Supervised by

Dr. Faisal Cheema

Co-Supervised by

Ms. Hira Mastoor



Department of Data Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

The Enhanced Driver Assist (EDA) system is designed to improve road safety by assisting drivers in making informed decisions and alerting them of potential hazards. The system integrates three primary modules: driving pattern analysis using OBD data, driver attention monitoring using an in-cabin live feed, and lane departure monitoring using a road camera feed. The project involves hardware and software components, combining edge computing with machine learning to provide timely assistance to drivers, particularly on highways.

1.1 Problem Statement

A substantial portion of traffic accidents (RTCs) globally are caused by driver fatigue and distraction. Fatigue-related rear-end collisions (RTCs) account for 54% of fatal accidents on motorways and 41% of serious injuries on national routes.[5] In Pakistan, 33% of incidents on the M2 motorway were caused by sleepy drivers.[8] The issue has been made worse by the dearth of dependable and reasonably priced technologies to track drivers' attentiveness and attentiveness. Furthermore, accidents resulting from lane departure occur when the driver is multi-tasking or distracted and doesn't realize that his car is going out of the lane, especially when driving at high speeds on highways. Hence, by adding some features like attention monitoring, lane departure detection, and pattern analysis, the EDA system seeks to address these problems by warning drivers of potentially dangerous driving styles. The voice alert of the system will increase traffic safety by warning the driver to stay focused and pay full attention to the road, avoiding any multitasking.

1.2 Scope

The scope of our Enhanced Driver Assist (EDA) project is to create a device that will monitor the driver's attention and distraction, will provide alert warnings in case of dangerous situation, and if the driver monitoring is not enough, it will also include warning beeps at lane departure, so if the driver is moving out of lane unintentionally or not, he will be alerted. We are targeting highways and motorways of Pakistan, since the lanes are marked there, and dangerous problems such as driver fatigue, distraction, and risky driving behaviors are rampant. The product is specifically designed for Locally assembled and popular cars, such as Toyota Corolla, Honda Civic, Honda City, etc, which can provide the minimum required OBD data for driving behavior analysis.

It will provide the following functionalities:

- Monitor the driver's eye or facial movements to detect any signs of drowsiness or fatigue.
- Lane detection and departure alert system which will alert the driver in case the car drifts out of lane.
- The system analyzes driving patterns through OBD-II data, identifying dangerous behaviors such as harsh acceleration or speeding.
- Voice alerts keep the driver focused by offering hands-free alerts and warnings.
- Dashcams and hardware work together to monitor both the driver and road conditions, with local processing for good response times.

1.3 Modules

The EDA system is divided into several modules, each responsible for specific functions of the overall driver assistance system. Each module is designed to integrate seamlessly with others, ensuring the system functions as a cohesive unit.

1.3.1 Module 1: Driver Attention Monitoring

This module will use the feed from the dashcam footage, and use it to focus on the driver's facial expressions and see if the driver is looking drowsy, or tired. Hence, when the detection is done, the EDA will alert the car driver and prevent any unforeseen accidents. [3] in this they share some ways to make compact and efficient models for less powerful devices. [7] this paper shares more techniques and possible ways for quantization

1. Facial landmarks and feature tracking using computer vision.
2. Drowsiness detection through eye movements.[3]
3. Facilitate User with an efficient alert warning, enhancing their driving experience.
4. For the dataset, we will use driver data from roboflow [2] and many more

1.3.2 Module 2: Lane Detection and Departure Warning

In this module, we will use the road-oriented dashcam footage where we will detect the lane markings on the highway and after detection, we will ensure that the car of the driver stays within its lane. In case of drifting outside of the lane, they will be alerted with either a beep or voice command, ensuring that the vehicle remains within lane boundaries.[4, 9] in this paper they use segmentation to check vehicle positioning.

1. Lane detection using camera input feed via OpenCV.
2. Alert system warning in case of lane departure detection.
3. System will work for lane-marked roads, like Highways and Motorways.
4. Dataset used for lane detection is tuSimple[1] and many more

1.3.3 Module 3: Driving Pattern Monitoring

Now for this module tracks driver behavior using OBD-II sensor data to monitor risky driving practices like harsh braking, speeding, and rapid acceleration.[6, 11] Using some of the basic data from obd (e.g speed, rpm, throttle position) we can not only monitor the drivers' behaviour but also improve it. The system provides feedback to promote safer driving habits.

1. Collection of driving data from OBD-II sensors.
2. Analysis of driving patterns (e.g., rash driving, speeding).
3. Alerts for unsafe driving practices.
4. Dataset will be collected locally on medium-sedan cars

1.3.4 Module 4: Voice Alert

This module enables hands-free interaction with the system. The voice alert can talk to the driver and warn them about various scenarios, prohibiting them from multitasking and helping the driver stay focused on the road.

1. Delivers voice alerts for potential hazards and driver assistance.
2. Processes alerts locally on edge devices for minimal response time and smooth interaction.

1.3.5 Module 5: Hardware Implementation and Integration

This module focuses on the integration of hardware components such as dashcams (in-cabin and out-cabin), Raspberry Pi or Nvidia Jetson Nano, Coral TPU, and OBD-II sensors.[10] This article explains some of the advantages of different hardware and their strong points. It ensures the synchronization of hardware and software for optimal performance. However, these components are subject to changes in the future with the availability.

1. Integration of OBD-II sensor for vehicle data collection.
2. Streaming videos from cameras to the models
3. Running lightweight models locally on the device
4. Synchronization between hardware and software components.

1.4 User Classes and Characteristics

User Class		Description
Driver		The driver is the primary user of the EDA system. The system assists drivers by providing timely alerts for drowsiness, distraction, lane departure, and driving behavior. Drivers are expected to engage with the system frequently while on highways or motorways. Feedback will be delivered via audio alerts, minimizing distractions. The driver may also view log reports about their driving performance, which will help them improve their driving habits.
Fleet	Man- ager	The fleet manager oversees multiple vehicles equipped with the EDA system. They will be responsible for monitoring driving patterns and driver attentiveness through periodic reports generated by the system. The system will provide reports or aggregated data, focusing on events such as frequent lane departures, harsh speeding, or prolonged drowsiness.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

Event ID	Event	Condition/Trigger	Response
EDA_OBD_01	Data received from OBD-II	Valid OBD-II data received (RPM, Speed, etc.)	Store data, trigger driver behavior analysis
EDA_OBD_02	Invalid OBD-II data	OBD-II sends corrupted/incomplete data	Generate error log, request resend, notify user
EDA_DROWS_01	Drowsiness detected	In-cabin video detects signs of fatigue (e.g. eyes closing)	Trigger alert to driver, sound alarm, log event
EDA_LANE_01	Lane departure detected	Vehicle crosses lane boundary without signaling	Trigger visual/audio alert, corrective steering
EDA_ALERT_01	timely driver alert	High-risk event detected (e.g. drowsiness, lane departure)	Send alert with minimum lag to driver, log event
EDA_ML_01	Unusual driver pattern	ML pattern analysis detects abnormal behavior	Trigger alert to driver, log the event, suggest corrective measures
EDA_ALERT_03	Alert system error	System fails to notify driver	Notify system admin, log error, attempt system restart

Table 2.1: Event-Response Table for Enhanced Driver Assist System

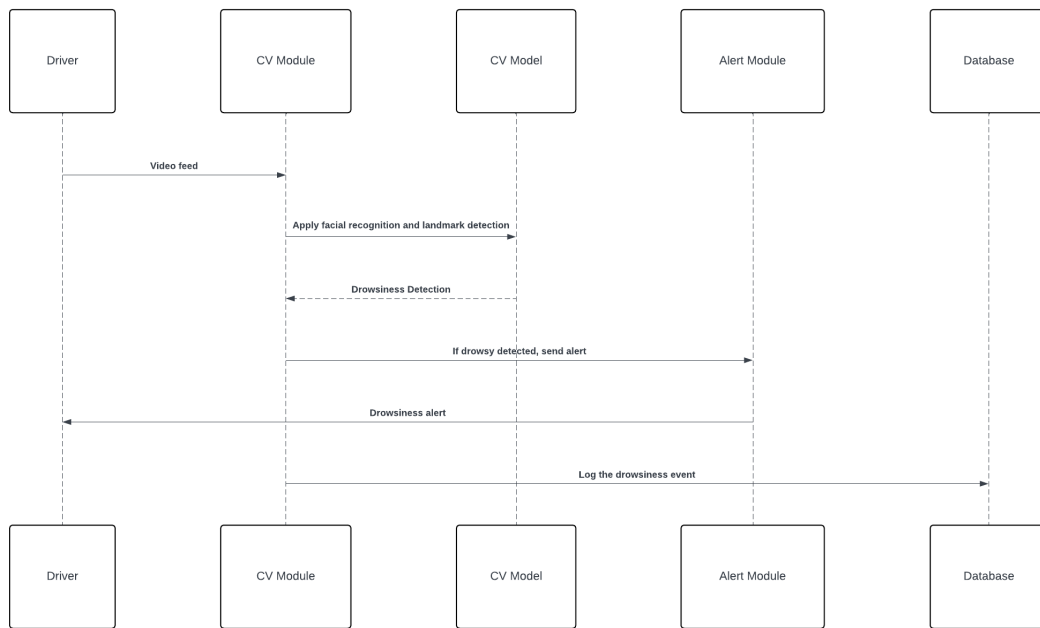


Figure 2.1: Driver Monitoring Sequence Diagram

2.2 Functional Requirements

Following are the functional requirements for each module:

2.2.1 Driver Attention Monitoring

Following are the requirements for Driver Attention Monitoring:

1. The system shall detect driver drowsiness based on eye closure duration.
2. The system shall issue an alert if the driver's attention is diverted for more than 5 seconds.

2.2.2 Lane Departure Monitoring

Following are the requirements for Lane Departure Monitoring:

1. The system shall detect lane markers on highways and monitor the vehicle's position.
2. The system shall warn the driver when the vehicle departs from its lane.

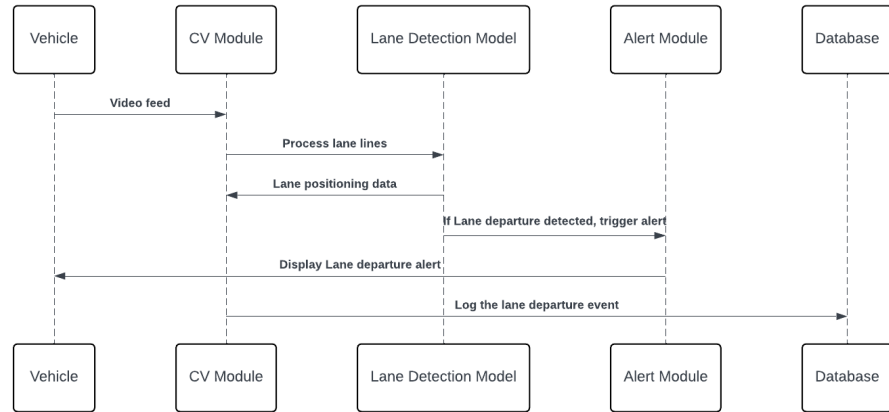


Figure 2.2: Lane Departure Sequence Diagram

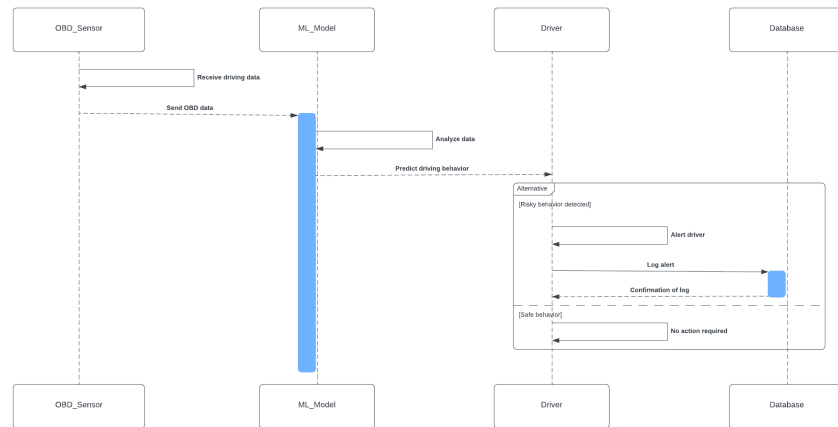


Figure 2.3: Driving Pattern Analysis Sequence Diagram

2.2.3 Driving Pattern Analysis

Following are the requirements for Driving Pattern Analysis:

1. The system shall analyze OBD-II data for speed, acceleration, and rpm.
2. The system shall generate reports for unsafe driving behavior, such as harsh acceleration or over-speeding.

2.2.4 Voice Alert

Following are the requirements for the Voice Alert module:

1. The system shall deliver voice alerts for potential hazards and driver assistance.

2. The system shall process alerts locally on edge devices to ensure minimal response time and smooth interaction.

2.2.5 Hardware Implementation and Integration

Following are the requirements for Hardware Implementation and Integration:

1. The system shall integrate OBD-II sensors for vehicle data collection.
2. The system shall stream videos from in-cabin and out-cabin cameras to the models.
3. The system shall run lightweight models locally on the device.

2.3 Non-Functional Requirements

2.3.1 Reliability

The system shall maintain a Mean Time Between Failures (MTBF) of at least 1000 hours. A failure is defined as any occurrence that prevents the system from performing its intended function. Consequences of software failure may include the inability to provide timely driver assistance and potential safety risks. To protect against failures, the system shall perform regular self-checks. A strategy for error detection will involve logging errors and monitoring system performance metrics, while error correction shall include alerts to the administrator.

2.3.2 Usability

The system shall provide an intuitive user interface that allows users to learn and operate the system with minimal training.

- USE-1: The EDA system shall allow a driver to access driving behavior feedback.
- USE-2: The system shall enable fleet managers to generate reports on driving patterns.

2.3.3 Performance

The EDA system shall meet the following performance requirements:

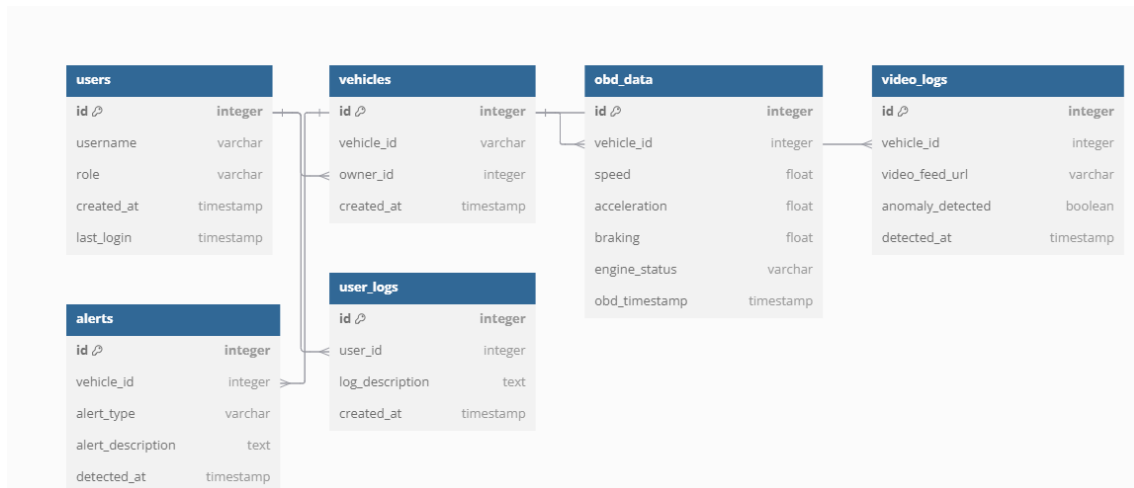


Figure 2.4: Domain Model of EDA System

- PER-1: The system shall analyze OBD-II data and provide driving behavior feedback within 4 seconds of data collection.
- PER-2: The voice alerts will try to be processed and delivered to the driver within 4 seconds of detection of a potential hazard.
- PER-3: most of lane detection warnings shall be issued with an approximate accuracy of 80% or higher.

2.3.4 Security

The system shall ensure that user data is protected against unauthorized access and breaches.

- SEC-1: The system shall limit access to sensitive data to authorized personnel only, based on predefined roles.
- SEC-2: The system shall log all access attempts to sensitive data for auditing purposes.

2.4 Domain Model

the figure 2.4 represents the domain model

Chapter 3

System Overview

The Enhanced Driver Assist (EDA) system is designed to improve road safety and driver performance through advanced monitoring and feedback mechanisms. The system utilizes a combination of hardware and software components, including in-cabin and out-cabin cameras, OBD-II sensors, and data processing capabilities to assess driver behavior, monitor lane positions, and provide timely alerts for hazardous situations.

This system integrates machine learning models to analyze driving patterns, drowsiness detection, and lane departure monitoring, ensuring that drivers receive immediate feedback to enhance their driving experience and safety. The system is particularly suited for highway driving, focusing on mid-range sedan cars like Toyota Corolla and Honda Civic.

3.1 Architectural Design

The EDA system is modular, comprising several interconnected components that work collaboratively to deliver the desired functionality. The main modules include:

- **Driver Attention Monitoring:** This module assesses driver alertness and provides warnings if drowsiness or distractions are detected.
- **Lane Departure Monitoring:** This module detects lane boundaries and warns the driver when the vehicle deviates from its lane.
- **Driving Pattern Analysis:** This module analyzes OBD-II data for driving behavior, generating reports on unsafe driving patterns.
- **Voice Alert:** This module provides voice notifications for potential hazards and driver assistance, helping to keep the driver focused.

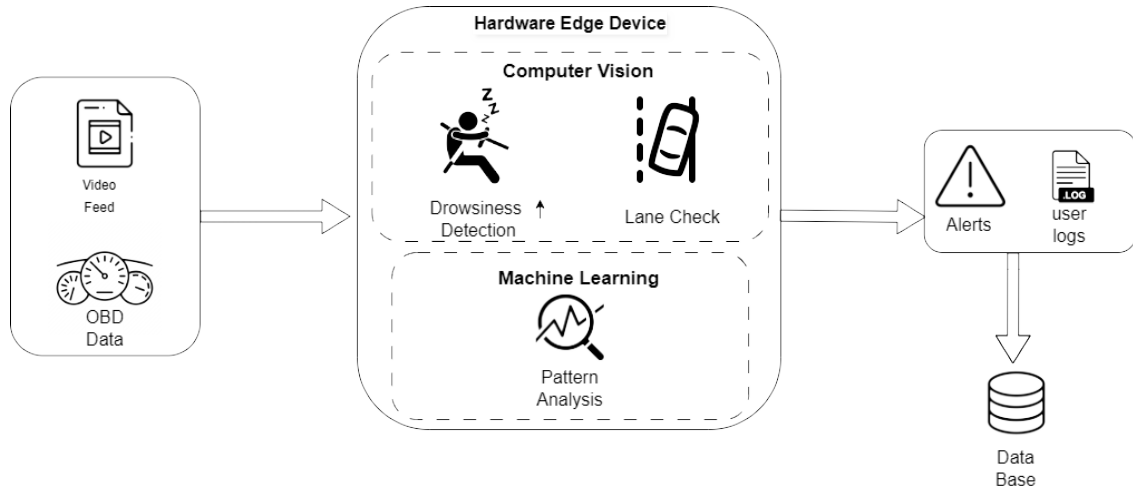


Figure 3.1: Architecture Diagram

- **Hardware Implementation and Integration:** This module handles the integration of various hardware components and ensures synchronization between hardware and software.

The system is designed using a layered architecture pattern that allows each module to function independently while communicating with others as needed. The data flows between the modules, ensuring a seamless experience for the user and maintaining system efficiency.

3.2 Design Models

The design models applicable to the EDA system, utilizing an object-oriented development approach, include the following:

- **Activity Diagram:** This diagram outlines the workflow and processes in the system. shown in figure 3.2
- **Data Flow Diagram (DFD):** This diagram provides a detailed view of the data flow within the system, outlining processes, data sources/sinks, and storage mechanisms, shown in figure 3.3
- **System-level Sequence Diagram:** This diagram shows the interactions between system components over time, focusing on the overall system behavior. Figure shown in 3.4

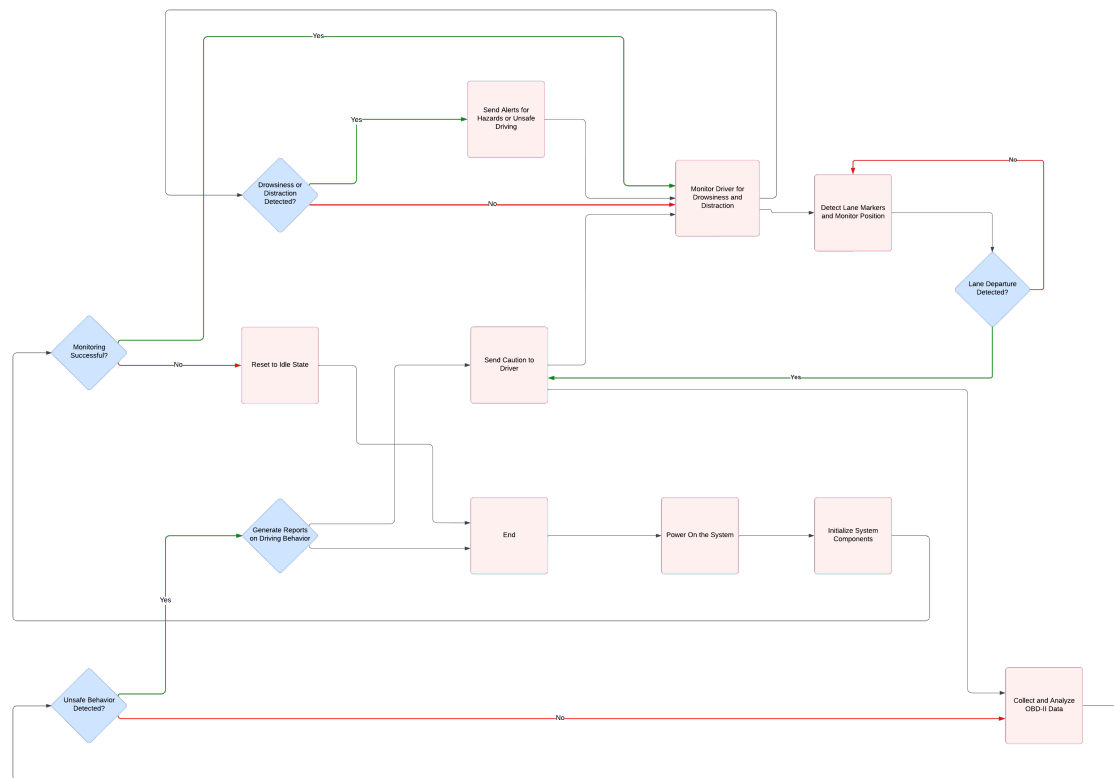


Figure 3.2: Activity Diagram

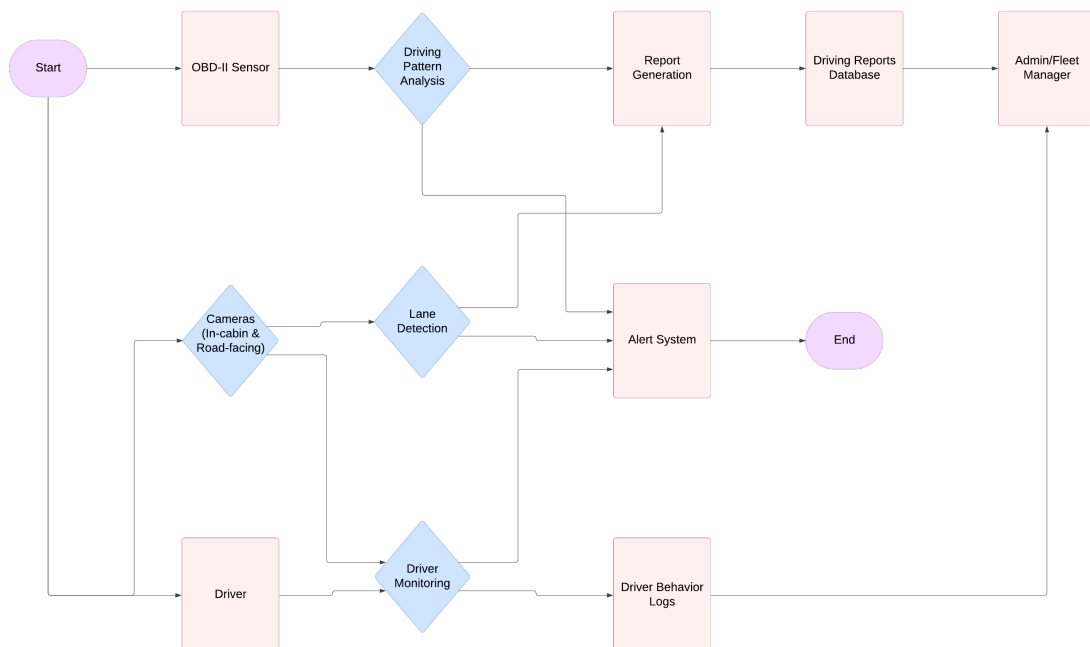


Figure 3.3: Data Flow Diagram

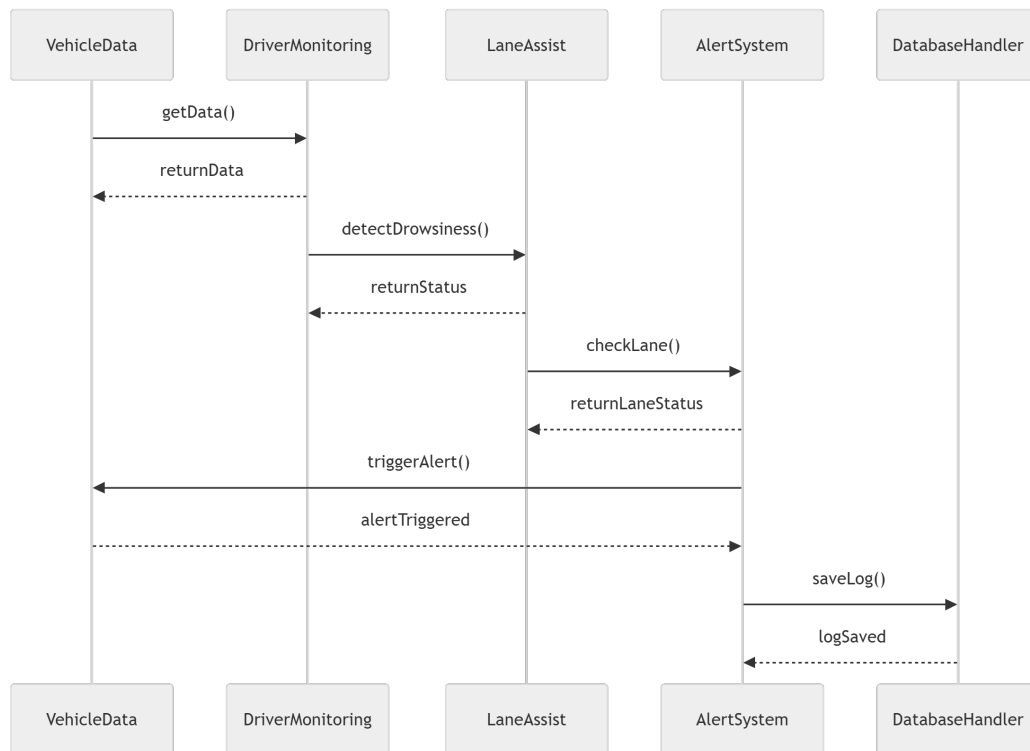


Figure 3.4: System-level Sequence Diagram

3.3 Data Design

The Enhanced Driver Assist (EDA) system handles a variety of data entities critical to its functionality. This section describes how the information within the system is transformed into data structures, processed, and stored using MongoDB Atlas as the primary database solution. Each data entity is organized to optimize storage, retrieval, and processing, ensuring the system's overall performance and reliability.

3.3.1 Data Entities

The major data entities in the EDA system include:

- **Driver Information:** Contains driver-specific details such as name, ID, vehicle type, driving history, and relevant profiles.
- **Logs:** Stores continuous logs from the vehicle, including OBD-II sensor data, timestamps, and driving patterns for analysis.
- **Alerts:** Records alerts generated by the system, such as drowsiness detection, lane departure warnings, and hazard alerts, along with timestamps and context of the alert.

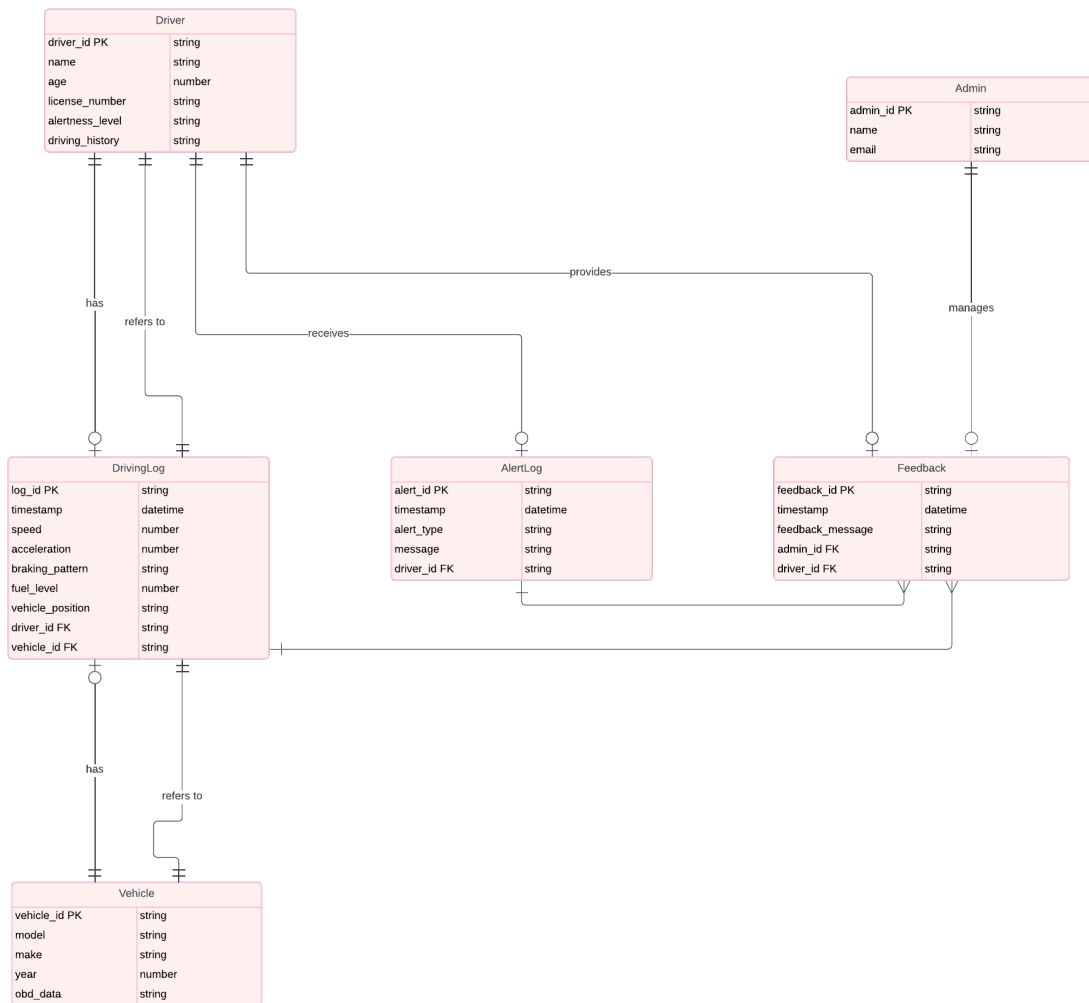


Figure 3.5: ERD

- **Timestamps:** All logs and alerts are tagged with precise timestamps for traceability and to allow accurate time-based analysis of driving behavior.

3.3.2 Data Storage and Processing

MongoDB Atlas is used as the cloud-hosted NoSQL database, providing a flexible and scalable solution for storing the unstructured data generated by the EDA system. Key considerations include:

- **Driver Information Collection:** This data is stored in a collection called ‘drivers’, where each document contains driver-specific details and metadata.
- **Logs:** Continuous OBD-II data logs are stored in a ‘logs’ collection. Each document includes a reference to the associated driver, vehicle, and timestamp to allow for pattern recognition and performance tracking.
- **Alerts:** Generated alerts are stored in an ‘alerts’ collection, tagged with event data such as the type of alert, associated driver, and the system’s response.
- **Feedback:** All logs and alerts are compiled for each driver into a report providing feedback and allowing accurate time-based analysis of driving behavior.
- **Admin:** can view reports of logs and alerts in the feedback.
- **vehicle:** contains vehicle information (which would preferably be a sedan according to domain specified).

This data design allows the EDA system to handle large volumes of diverse data efficiently, ensuring scalability and performance in driver assistance scenarios.

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